

SeungHun Lee

+82 (0)10-7495-7445 • Mr.leesh90@gmail.com • <https://leesh90.github.io/about/>

Professional Experience

3billion, Seoul, South Korea

AI Engineer

March 2026 - Present

- Developing foundation models for DNA, RNA, and protein sequences.
- Applying deep learning architectures for large-scale biological sequence modeling.
- Building scalable training pipelines for large-scale genomic datasets.

Current Research Interests

My research focuses on AI-driven modeling of biological sequences, particularly in genomics and molecular biology. I develop foundation models to understand genomic function and predict the impact of genetic variants. I am particularly interested in large-scale sequence modeling and representation learning for DNA/RNA/protein data, along with explainable AI (XAI) techniques to improve model interpretability in high-stakes domains.

• Education

• Sangmyung University, Cheonan, South Korea (Advisor: Prof. Hyun-chul Kim)

- Ph.D., Computer Software Engineering, GPA: 4.45/4.5 March 2018 – Aug 2026
- M.S., Computer Software Engineering, GPA: 3.68/4.5 March 2015 - Feb 2017
- B.S., Computer Software Engineering, GPA: 4.24/4.5 March 2009 - Feb 2015

Doctoral Research

My doctoral research spanned digital finance, network security, and Explainable AI (XAI). In digital finance, I employ a multimodal approach (i.e., text, video, and audio) and user behavior data to analyze crowdfunding platforms, detecting fraudulent campaigns and identifying success factors. For network traffic, I focus on traffic classification and intrusion detection, analyzing both traffic headers and payloads while developing methods to classify traffic patterns. Across these domains, I leverage advanced AI methodologies including natural language processing, pattern analysis, classical machine learning algorithms, transformers, graph neural networks (GNNs), large language models (LLMs), and Agent AI systems. Throughout my work, I integrate XAI techniques to enhance the reliability, transparency, and interpretability of AI-driven solutions.

Publications

1. Seunghun Lee, Muneeb A. Khan, and Hyun-Chul Kim. "Explainable Prediction of Crowdfunding Success Using Hierarchical Attention Network." MDPI electronics, vol. 15, no. 3, Jan. 2026.
2. Seunghun Lee, Heemin Park, and Hyun-Chul Kim. "Fraud Detection on Crowdfunding Platforms Using Multiple Feature Selection Methods." IEEE Access, Mar. 2025.
3. Seunghun Lee, Wafa Shafqat, and Hyun-chul Kim. "Backers beware: Characteristics and detection of fraudulent crowdfunding campaigns." Sensors, vol. 22, no. 19, Oct. 2022.
4. Kang-Hee Lee, Seunghun Lee, and Hyun-Chul Kim. "Traffic classification using deep learning: being highly accurate is not enough." In Proceedings of the SIGCOMM, Aug. 2020. (poster)
5. Seunghun Lee, KangHee Lee, and Hyun-chul Kim. "Content-based success prediction of crowdfunding campaigns: A deep learning approach." In Proceedings of the ACM conference on Computer Supported Cooperative Work and Social Computing (CSCW), Nov. 2018. (poster)
6. Seunghun Lee, and Hyun-chul Kim. "Crowdfunding Scams: The Profiles and Language of Deceivers." Journal of The Korea Society of Computer and Information, vol. 23, no.3, Mar. 2018.

7. Kang-hee Lee, Seunghun Lee, and Hyun-chul Kim. "Predicting Success of Crowdfunding Campaigns using Multimedia and Linguistic Features." *Journal of Korea Multimedia Society*, vol. 21, no. 2, Feb. 2018.
8. Wafa Shafqat, Seunghun Lee, Sehrish Malik, and Hyun-chul Kim. "The language of deceivers: Linguistic features of crowdfunding scams." In *Proceedings of the International Conference Companion on World Wide Web(WWW)*, Apr. 2016 (poster)
9. Yoon, Jangho, Seunghun Lee, and Hyun-chul Kim. "Smart SNS Map: Location-based Social Network Service Data Mapping and Visualization System." *Journal of Korea Multimedia Society*, vol. 19, no. 2, Feb. 2016.
10. Seunghun Lee, Daeyoung Oh, Minhyuk Kang, and Hyun-chul kim. "SNS Map: Location-based SNS data mapping system." In *Proceedings of the Korea Computer Congress (KCC)*, June. 2015.

Projects

Explainability in Graph Neural Networks for Internet Traffic Classification June 2023 – Feb 2025

- Trained and classified encrypted traffic using GNNs on host behavior and network-wide graphs.
- Applied graph XAI methods (e.g., GNNExplainer) to reveal distinctive graph structural patterns.

Research on Virtualization-based 5G Networks and Cyber Threats April 2023 – Oct 2023

- Developed Cyber5Gym, a virtualization-based multi-agent training environment for 5G networks using Open5GS and UERANSIM.
- Implemented cyber threat scenarios including high-bandwidth traffic, SMC replay, and DDoS attacks.

Deep Learning based Internet Traffic Classification: Myths, Realities, and their Explainabilities June 2022 – May 2023

- Applied Hierarchical Attention Networks (HAN) to Internet traffic classification.
- Used attention visualization to explain and identify key traffic features.

Research on Building a Virtualization-Based 5G Cyber Training Environment April 2022 – Oct 2022

- Built a virtualization-based 5G cyber training environment using Open5GS and UERANSIM.
- Investigated major 5G vulnerabilities, detailing attack procedures, impacts, and countermeasures.

Multi-modal data-driven Explainable AI Systems and the Future of Digital Finance Sep 2019 – Feb 2022

- Analyzed text using HAN with attention visualization for feature interpretation.
- Applied CNNs and saliency maps for video and image analysis.
- Extracted acoustic features via MFCC and modeled metadata with MLP.
- Integrated all modalities into a unified multi-modal XAI system for fraud detection in digital finance.

Towards Explainable AI in Next-Generation Intrusion Detection Systems April 2019 – Oct 2019

- Applied HAN to network traffic classification using its hierarchical flow–packet–byte structure.
- Used attention visualization to identify salient header fields and payload patterns.

Statistics-based Network Behavior Modeling May 2018 – Oct 2018

- Modeled network behavior using host analysis and Latent Dirichlet Allocation (LDA) for anomaly detection.
- Analyzed and visualized anomalous patterns via host behavior graphs and radar charts.

Traffic Measurement in Anonymity Networks April 2017 – Oct 2017

- Collected and analyzed network traffic from the Tor anonymity network.

Characterization and Automatic Labeling of Malicious Traffic in Control System Networks April 2017 – Oct 2017

- Applied Latent Dirichlet Allocation (LDA) to classify Industrial Control System (ICS) network traffic.
- Modeled traffic flows as documents to automatically extract signatures and label unclassified SCADA traffic.

Network Traffic Classification for Intrusion Detection

June 2015 – Dec 2015

- Applied LDA for automated signature detection and classification of general application network traffic.

Honors and Awards

Teaching Assistant Scholarship, Computer Software Engineering, Sangmyung Univ.	2019 - 2020
Research Assistant Scholarship, Computer Software Engineering, Sangmyung Univ.	2015 – 2016
Second Honors Graduate, B.S., Computer Software Engineering, GPA: 4.24/4.5	2015
Award for Creative Design Competition (Graduation Project), College of Engineering	2014

Teaching Experience

Introduction to Data Science (Big Data), Teaching Assistant and Lab Instructor	2019 – 2025
Algorithms, Teaching Assistant	2018 – 2024
Computer Programming, Teaching Assistant and Lab Instructor	2017, 2019